

WaySense: Infrastructure-Light Indoor Navigation

Designing a scalable localization engine for GPS-denied environments by repurposing existing Wi-Fi infrastructure.

My Role:

- **System Architect:** Defined the sensor topology and selected the wireless protocol (2.4GHz Wi-Fi) based on signal propagation analysis.
- **Interaction Designer:** Designed the "Context Handover" protocol to manage user expectations during the transition from GPS to Local tracking.
- **IoT Lead:** Orchestrated the integration of Occupancy Sensors with the Fingerprinting Database.

Tech Stack: Wi-Fi RSSI Fingerprinting, Python (Backend), SQL, Infrared Sensors, TP-Link Hardware.

1. The Design Problem: The "Blind Handoff"

GPS is the global standard for navigation, but it has a critical failure mode: **Signal occlusion**. When a vehicle enters a concrete structure (like a basement or parking complex), GPS accuracy drops from <5m to "No Signal."

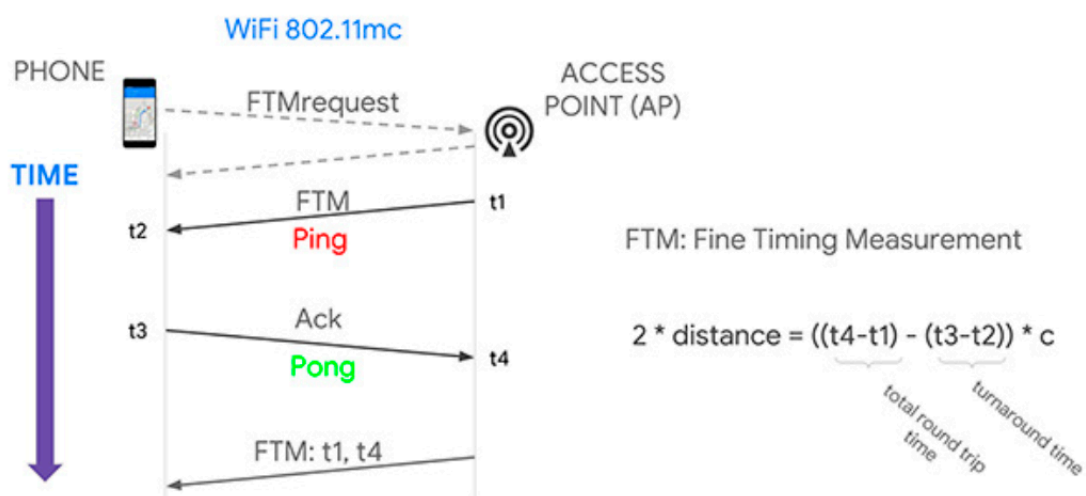
This creates a **"Blind Handoff"**—a jarring user experience where the driver is suddenly left without guidance in a complex, high-stress environment.

- **The Constraint:** Deploying proprietary hardware (LiDAR, UWB, or BLE Beacons) is prohibitively expensive for large-scale public infrastructure.
- **The Challenge:** *How might we achieve reliable indoor localization using only the hardware that already exists (Wi-Fi)?*

2. The Engineering Logic: Managing Trade-offs

Decision A: RSSI Fingerprinting vs. RTT (Cost vs. Compatibility)

- **The Analysis:** I evaluated **Wi-Fi RTT (Round Trip Time)**, which offers superior precision (1-2m). However, it requires specific hardware (Android 9+ and expensive RTT-enabled APs).



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- **The Decision:** I chose **RSSI Fingerprinting** (5m precision).
- **The Rationale:** While less precise, RSSI is universal. By using standard **TP-Link EAP225** Access Points, We prioritized **Accessibility** and **Scalability** over raw precision, ensuring the system works on any smartphone without custom hardware.

Decision B: Spectrum Selection

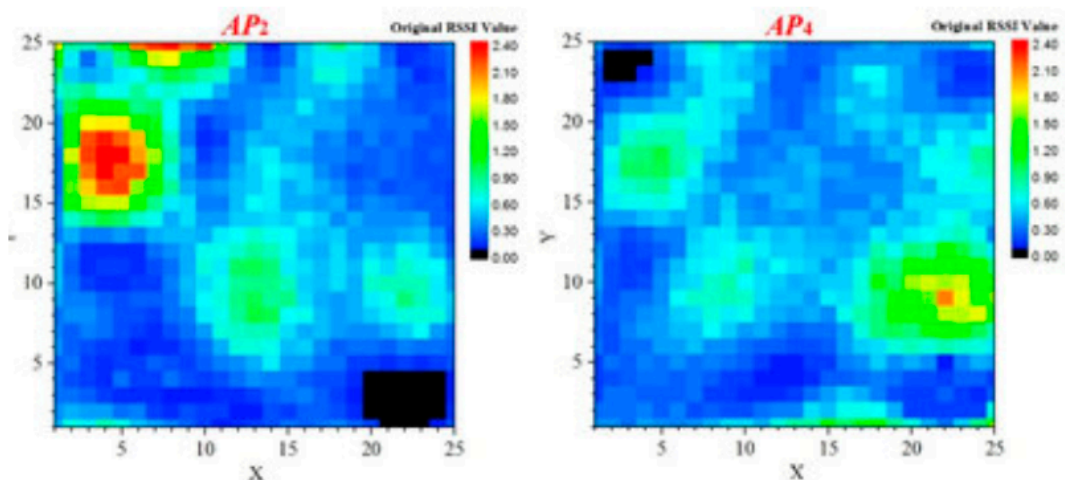
- **The Analysis:** 5GHz Wi-Fi offers speed but suffers from high attenuation through concrete walls.
- **The Decision:** I constrained the system to the **2.4GHz spectrum**.
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- **The Rationale:** This maximized signal range (20-100m) and wall penetration, reducing the number of APs required to cover the floor plan—a critical "Design for Manufacturing/Assembly" (DFMA) consideration .

3. The Digital Twin Approach

I treated the invisible signal environment as a topographic map.

Intervention A: The Signal Heatmap

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- **The Action:** We mapped the facility by recording RSSI values at reference points, storing them in a **Fingerprinting Database** .
- **The Visual:** I visualized this data as a **Signal Heatmap**, creating a "Digital Twin" of the radio environment. This allowed us to identify "Dead Zones" and reposition Access Points for optimal coverage.



Intervention B: The Context-Aware Handover

- **The Action:** The system detects the specific "Signal Drop" (**RSSI Threshold Hysteresis**) pattern of entering the facility.
- **The Design:** Instead of showing a "GPS Signal Lost" error, the App automatically switches context to the "**Floor Plan View**," seamlessly handing over control to the local Wi-Fi grid.

4. The System Architecture

diagram code -> "sequenceDiagram

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autonumber
participant User as Driver
participant App as WaySense App (Frontstage)
participant Server as Backend System (Backstage)
participant Sensor as IoT Sensors

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Note over User, Sensor: PHASE 1: THE HANDOVER

User->>App: Enters Basement (GPS Signal Dies)

App->>Server: Detects GPS Loss + Connects to Wi-Fi

Server-->>App: "Context Switch" Triggered

App->>User: Show Notification: "Switched to Indoor Mode"

Note over User, Sensor: PHASE 2: NAVIGATION

User->>App: Moves forward 10 meters

App->>Server: Sends RSSI Signal Strength (-65dBm)

Server->>Server: Triangulate Position via Fingerprint DB

Server-->>App: Returns Vector Path (x,y coordinates)

App->>User: Updates Map UI (Real-time Blue Dot)

Note over User, Sensor: PHASE 3: ARRIVAL

User->>Sensor: Parks in Spot #B24

Sensor->>Server: IR Sensor Detects Object

Server->>Server: Update Parking DB to "Occupied"

Server-->>App: Send "Arrival" Confirmation

App->>User: "You have arrived!"

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The Loop:

1.

Input: Smartphone reads ambient RSSI + Occupancy Sensors detect vacancy .

2.

Logic: Server triangulates position using the Fingerprint DB + Calculates shortest path to an *Open spot* .

3. **Output:** User receives a dynamic vector path.

graph LR

%% STYLING

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classDef sensor fill:#fff3e0,stroke:#ef6c00,stroke-width:2px;

%% NODES

subgraph Frontstage [USER EXPERIENCE]

direction TB

User("USER (Smartphone)
<i>Captures RSSI Signal</i>"):::user

end

subgraph Infrastructure [PHYSICAL LAYER]

direction TB

AP("WI-FI ACCESS POINTS
<i>Signal Beacon</i>"):::infra

Sensor("IOT SENSORS
<i>Occupancy Detection</i>"):::sensor

end

subgraph Backstage [SYSTEM LOGIC]

direction TB

Server("NAVIGATION ENGINE
<i>Triangulation Algorithm</i>"):::system

DB[("FINGERPRINT DB
<i>Digital Twin Map</i>")]:::system

end

%% FLOW

AP --> User

User -- "1. Raw Signal Data" --> Server

Sensor -- "2. Spot Availability" --> Server

Server <--> DB

Server -- "3. Vector Path (x,y)" --> User

%% LINK STYLE

linkStyle default stroke:#333,stroke-width:1.5px;

Fig 3. System Architecture: The smartphone acts as a sensor, capturing ambient Wi-Fi signals. The backend engine cross-references this real-time data against a "Fingerprint Database" to generate coordinates, while simultaneously pulling vacancy data from IoT sensors to calculate the optimal path.

5. Validation & Impact

- **Precision:** Achieved **~5m accuracy** using standard hardware, sufficient for vehicle wayfinding contexts where "Lane-level" precision is key.

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Scalability: Validated architecture for **200 concurrent users** per Access Point.

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Efficiency: Reduced hardware cost by **~90%** compared to BLE Beacon solutions.